Exercises 7.5

Applying l'Hôpital's Rule

Use l'Hôpital's rule to find the limits in Exercises 7–50.

42. $\lim_{x \to 0^+} (\csc x - \cot x + \cos x)$

Indeterminate Powers and Products

Find the limits in Exercises 51–66.

55. $\lim_{x \to 0^+} x^{-1/\ln x}$

57. $\lim_{x \to \infty} (1 + 2x)^{1/(2 \ln x)}$

77. Only one of these calculations is correct. Which one? Why are the others wrong? Give reasons for your answers.

a. $\lim_{x \to 0^+} x \ln x = 0 \cdot (-\infty) = 0$

b. $\lim_{x \to 0^+} x \ln x = 0 \cdot (-\infty) = -\infty$

c. $\lim_{x \to 0^+} x \ln x = \lim_{x \to 0^+} \dfrac{\ln x}{(1/x)} = \dfrac{-\infty}{\infty} = -1$

d. $\lim_{x \to 0^+} x \ln x = \lim_{x \to 0^+} \dfrac{\ln x}{(1/x)}$

$$= \lim_{x \to 0^+} \frac{(1/x)}{(-1/x^2)} = \lim_{x \to 0^+} (-x) = 0$$

85. Show that

$$\lim_{k\to\infty}\left(1+\frac{r}{k}\right)^k = e^r.$$

Exercises 7.6

L'Hôpital's Rule

Find the limits in Exercises 91–98.

92. $\lim_{x\to 1^+} \dfrac{\sqrt{x^2-1}}{\sec^{-1}x}$

97. $\lim_{x\to 0^+} \dfrac{\left(\tan^{-1}\sqrt{x}\right)^2}{x\sqrt{x+1}}$

Exercises 7.8

Big-oh and Little-oh; Order

9. True, or false? As $x \to \infty$,

a. $x = o(x)$

b. $x = o(x + 5)$

c. $x = O(x + 5)$

d. $x = O(2x)$

e. $e^x = o(e^{2x})$

f. $x + \ln x = O(x)$

g. $\ln x = o(\ln 2x)$

h. $\sqrt{x^2 + 5} = O(x)$

11. Show that if positive functions $f(x)$ and $g(x)$ grow at the same rate as $x \to \infty$, then $f = O(g)$ and $g = O(f)$.

20. The function e^x outgrows any polynomial Show that e^x grows faster as $x \to \infty$ than any polynomial

$$a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0.$$

22. The function ln x grows slower than any polynomial Show that $\ln x$ grows slower as $x \to \infty$ than any nonconstant polynomial.

Chapter 7 Practice Exercises

Evaluating Integrals

Evaluate the integrals in Exercises 31–78.

78. $\displaystyle\int \frac{dt}{(3t+1)\sqrt{9t^2+6t}}$

Applying L'Hôpital's Rule

Use l'Hôpital's Rule to find the limits in Exercises 85–108.

105. $\displaystyle\lim_{t\to 0^+}\left(\frac{e^t}{t}-\frac{1}{t}\right)$

112. True, or false? Give reasons for your answers.

a. $\dfrac{1}{x^4} = O\left(\dfrac{1}{x^2}+\dfrac{1}{x^4}\right)$ **b.** $\dfrac{1}{x^4} = o\left(\dfrac{1}{x^2}+\dfrac{1}{x^4}\right)$

c. $\ln x = o(x+1)$ **d.** $\ln 2x = O(\ln x)$

e. $\sec^{-1} x = O(1)$ **f.** $\sinh x = O(e^x)$

Chapter 7 Additional and Advanced Exercises

Limits

Find the limits in Exercises 1–6.

2. $\lim_{x\to\infty} \frac{1}{x}\int_0^x \tan^{-1} t\, dt$

5. $\lim_{n\to\infty}\left(\frac{1}{n+1}+\frac{1}{n+2}+\cdots+\frac{1}{2n}\right)$

16. Let g be a function that is differentiable throughout an open interval containing the origin. Suppose g has the following properties:

i) $g(x+y)=\dfrac{g(x)+g(y)}{1-g(x)g(y)}$ for all real numbers x, y, and $x+y$ in the domain of g.

ii) $\lim_{h\to 0} g(h)=0$

iii) $\lim_{h\to 0} \dfrac{g(h)}{h}=1$

a. Show that $g(0)=0$.

b. Show that $g'(x)=1+[g(x)]^2$.

c. Find $g(x)$ by solving the differential equation in part (b).

Exercises 8.1

Assorted Integrations

The integrals in Exercises 1–40 are in no particular order. Evaluate each integral using any algebraic method or trigonometric identity you think is appropriate, and then use a substitution to reduce it to a standard form.

33. $\displaystyle\int_{-1}^{0}\sqrt{\frac{1+y}{1-y}}\,dy$

47. The functions $y = e^{x^3}$ and $y = x^3e^{x^3}$ do not have elementary antiderivatives, but $y = (1 + 3x^3)e^{x^3}$ does.

Evaluate

$$\int (1 + 3x^3)e^{x^3}dx.$$

48. Use the substitution $u = \tan x$ to evaluate the integral

$$\int \frac{dx}{1 + \sin^2 x}.$$

49. Use the substitution $u = x^4 + 1$ to evaluate the integral

$$\int x^7\sqrt{x^4 + 1}\,dx.$$

Exercises 8.3

Integrating Square Roots

Evaluate the integrals in Exercises 23–32.

24. $\displaystyle\int_0^{\pi} \sqrt{1 - \cos 2x}\,dx$

28. $\displaystyle\int_0^{\pi/6} \sqrt{1 + \sin x}\, dx$

$\left(\textit{Hint:}\ \text{Multiply by } \sqrt{\dfrac{1 - \sin x}{1 - \sin x}}.\right)$

46. $\displaystyle\int_{-\pi/4}^{\pi/4} 6 \tan^4 x\, dx$

Applications

69. Arc length Find the length of the curve

$$y = \ln(\sec x), \quad 0 \le x \le \pi/4.$$

Exercises 8.4

Using Trigonometric Substitutions

Evaluate the integrals in Exercises 1–14.

12. $\displaystyle\int \frac{\sqrt{y^2 - 25}}{y^3}\, dy, \quad y > 5$

Assorted Integrations

Use any method to evaluate the integrals in Exercises 15–34. Most will require trigonometric substitutions, but some can be evaluated by other methods.

18. $\displaystyle\int \frac{dx}{x^2\sqrt{x^2+1}}$

26. $\displaystyle\int \frac{x^2\,dx}{(x^2-1)^{5/2}}, \quad x > 1$

In Exercises 35–48, use an appropriate substitution and then a trigonometric substitution to evaluate the integrals.

46. $\displaystyle\int \sqrt{\frac{x}{1-x^3}}\,dx$

(*Hint:* Let $u = x^{3/2}$.)